Joint inertial and actuation parameter estimation for the Space CoBot free-flyer

Rodrigo Ventura

January 22, 2024

Contents

1 System dynamics **1**

2 Parameter estimation **3**
2.1 Attitude subsystem 3
2.1.1 Discretization 4
2.1.2 Integral form 4
2.1.3 Single shooting 5
2.1.4 Multiple shooting 5
2.1.5 Variational integration approach 6
2.2 Linear algebraic approach 8

3 Gravity compensation **10**
3.1 Attitude subsystem 10
3.2 Full system 10

A Rough estimation of Space CoBot inertial parameterts **11**

1 System dynamics

Let $v \in \mathbb{R}^3$ and $\omega \in \mathbb{R}^3$ be the linear and angular velocities of a rigid body, $R \in \mathbb{R}^{3\times 3}$ be its attitude expressed as a rotation matrix, and $F \in \mathbb{R}^3$ and $M \in \mathbb{R}^3$ the force and torque in body frame. The motion of such a rigid body, when the center of mass is not coincident with the origin of the body

reference frame, is given by

$$M \begin{pmatrix} \dot{v} \\ \dot{\omega} \end{pmatrix} + C(\omega) = \begin{pmatrix} RF \\ M \end{pmatrix} \qquad \dot{R} = RS(\omega) \tag{1}$$

for

$$M = \begin{bmatrix} mI_{3\times3} & mS^T(c) \\ mS(c) & J_C + mS(c)S^T(c) \end{bmatrix} \tag{2}$$

and

$$C(\omega) = \begin{pmatrix} mS^2(\omega)c \\ S(\omega)\left[J_c + mS(c)S^T(c)\right]\omega \end{pmatrix} \tag{3}$$

where the inertial parameters are mass $m \in \mathbb{R}^+$, center of mass $c \in \mathbb{R}^3$, and moment of inertia $J_C \in \mathbb{R}^3$ with respect to the center of mass. The function S is the skew-symmetric operator, defined by

$$S(v) = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix} \tag{4}$$

where $v = [v_1\, v_2\, v_3]^T \in \mathbb{R}^3$.

The moment of inertia with respect to the body frame can be obtained using $J_B = J_C + mS(c)S^T(c)$, thus simplifying M into

$$M = \begin{bmatrix} mI_{3\times3} & mS^T(c) \\ mS(c) & J_B \end{bmatrix} \tag{5}$$

Since the Space CoBot free-flyer is mounted on a gimble, and assuming that the body frame is centered on the gimble rotation axis, the center of mass displacement c will cause a parasitic torque due to gravity. Let g be the gravity acceleration vector, $g = [0\,0\,-G]^T$ where $G \simeq 9.8m/s^2$ is the Earth gravity acceleration. Then, this torque, in body frame, is given by $M_g = c \times R^T g = S(c)R^T g$. Including the actuation, the net force and torque with respect to the body frame are given by

$$\begin{pmatrix} F \\ M \end{pmatrix} = Au + \begin{pmatrix} 0 \\ S(c)R^T g \end{pmatrix} \tag{6}$$

where $A \in \mathbb{R}^{6\times6}$ is the actuation matrix and $u \in \mathbb{R}^6$ is the actuation vector.

2 Parameter estimation

Dividing (1) by the mass m, and letting $C_1 = C/m$, $A_1 = A/m$, and $J_1 = J_C/m$, we obtain

$$\begin{pmatrix} R^T\left[\dot{v} + S^T(c)\dot{\omega} + S^2(\omega)c\right] \\ J_1\dot{\omega} + S(c)v + S(\omega)J_1\omega - S(c)R^T g \end{pmatrix} = A_1 u \tag{7}$$

The moment of inertia J_C is always symmetric, and thus, both J_B and J_1 are also symmetric. Therefore, J_1 can be written as

$$J_1 = LL^T \tag{8}$$

where $L \in \mathbb{R}^{3\times 3}$ is a lower triangular matrix. Then, the resulting model becomes

$$\begin{pmatrix} R^T\left[\dot{v} + S^T(c)\dot{\omega} + S^2(\omega)c\right] \\ LL^T\dot{\omega} + S(c)v + S(\omega)J_1\omega - S(c)R^T g \end{pmatrix} = A_1 u \tag{9}$$

2.1 Attitude subsystem

As a first step, consider the attitude dynamics alone. Consider $v = 0$ and let us separate the normalized actuation matrix A_1 in its translational and rotational parts, $A_{1F}, A_{1M} \in \mathbb{R}^{3\times 6}$

$$A_1 = \left[\frac{A_{1F}}{A_{1M}}\right] \tag{10}$$

The attitude dynamics in (9) simplifies to

$$LL^T\dot{\omega} + S(\omega)LL^T\omega - S(c)R^T g = A_{1M}u \tag{11}$$

Having collected a set of samples $\{(\dot{\omega}_k, \omega_k, R_k, u_k)\}$, the inertial parameters can be obtained from a constrained least-squares optimization

$$\begin{aligned} \text{minimize} \quad & \sum_k \left\| LL^T\dot{\omega}_k + S(\omega_k)LL^T\omega_k - S(c)R_k^T g - A_{1M}u_k \right\|^2 \\ \text{w.r.t.} \quad & L, c, A_{1M} \\ \text{s.t.} \quad & \|A_{1M}\|^2 = 1 \end{aligned} \tag{12}$$

where the constraint is a regularisation to prevent the trivial solution $\{L = 0, c = 0, A_{1M} = 0\}$.

Remark: section 2.2 presents an alternative approach to this optimization problem, where a solution can be computed in closed-form, since the cost function is squared norm of a linear function of the parameters. This observation simplifies enourmously the problem.

2.1.1 Discretization

Usually, $\dot{\omega}$ cannot be directly measured. One solution is discretizing system (11). Using Euler's implicit midpoint method[1], according to which a system $\dot{y} = f(y)$ is discretized into

$$y_{k+1} - y_k = hf(\frac{y_{k+1} + y_k}{2}) \tag{13}$$

for h being the time step, the discretization of (11) becomes

$$LL^T \Delta\omega_{k,k+1} + S(\omega_{k,k+1})LL^T\omega_{k,k+1} - S(c)R_{k,k+1}^T g = A_{1M}u_{k,k+1} \tag{14}$$

where $\Delta\omega_{k,k+1} = (\omega_{k+1} - \omega_k)/h$, $\omega_{k,k+1} = (\omega_{k+1} + \omega_k)/2$, $u_{k,k+1} = (u_{k+1} + u_k)/2$, and $R_{k,k+1}$ is the SO(3) average of rotation matrices R_k and R_{k+1} given by[2]

$$R_{k,k+1} = USV \tag{15}$$

where UDV is the SVD decomposition of $(R_k + R_{k+1})$ and $S = \mathrm{diag}(1, 1, |U||V|)$, for $|\cdot|$ denoting matrix determinant. The resulting least-squares problem is then

$$\begin{aligned} \text{minimize} \quad & \sum_k \left\| LL^T \Delta\omega_{k,k+1} + S(\omega_{k,k+1})LL^T\omega_{k,k+1} - S(c)R_{k,k+1}^T g - A_{1M}u_{k,k+1} \right\|^2 \\ \text{w.r.t.} \quad & L, c, A_{1M} \\ \text{s.t.} \quad & \|A_{1M}\|^2 = 1 \end{aligned} \tag{16}$$

Note that L is a lower (3×3) triangular matrix, and thus has only 6 unknowns. The total number of unknowns in this optimization problem is thus 27.

2.1.2 Integral form

Instead of discretizing (11), one could integrate over a short period of time, taking advantage that the rate of measurements of ω is typically very high (order of magnitude of 1kHz). Integrating (11) from t_k to t_{k+1}, one gets

$$LL^T(\omega_{k+1} - \omega_k) + \int_{t_k}^{t_{k+1}} S(\omega)LL^T\omega dt - S(c)\int_{t_k}^{t_{k+1}} R^T g dt = A_{1M}\int_{t_k}^{t_{k+1}} u dt \tag{17}$$

[1] A nice property of this simple colocation method is being symplectic, that is, it preserves energy and momentum when applied to Hamiltonian systems.

[2] Gramkow, C. (2001). On averaging rotations. Journal of Mathematical Imaging and Vision, 15(1-2), 7-16.

Assuming constant actuation from t_k to t_{k+1}, this simplifies to

$$LL^T(\omega_{k+1} - \omega_k) + \int_{t_k}^{t_{k+1}} S(\omega)LL^T\omega dt - S(c)\int_{t_k}^{t_{k+1}} R^T g dt = (t_{k+1} - t_k)A_{1M}u \tag{18}$$

where the integrals can be obtained ny numerically integrating the high rate angular velocity measurements.

2.1.3 Single shooting

An alternative way to perform parameter estimation is to simulate the whole interval and minimize the error between the measurements and the simulated trajectory.

Consider $N+1$ measurements of the attitude R_k, taken at time instants $t_0, \ldots, t_N$. Let $\Phi(R_0; t_0, t)$ be the solution of the attitude subsystem

$$\begin{aligned} LL^T\dot{\omega} &= -S(\omega)LL^T\omega + S(c)R^T g + A_{1M}u \\ \dot{R} &= RS(\omega) \end{aligned} \tag{19}$$

at time t given the initial condition R_0 at time t_0. The resulting optimization problem becomes

$$\begin{aligned} \text{minimize} \quad & \sum_{k>0} \psi\left(R_k, \Phi(R_0; t_0, t_k)\right) \\ \text{w.r.t.} \quad & L, c, A_{1M} \\ \text{s.t.} \quad & \|A_{1M}\|^2 = 1 \end{aligned} \tag{20}$$

where $\psi(R_\alpha, R_\beta)$ is a rotation error function, e.g.,

$$\psi(R_\alpha, R_\beta) = \frac{1}{2}\text{tr}\left(I - R_\alpha R_\beta^T\right) \tag{21}$$

Note that the actuation signal u must be the same for the real system and the attitude subsystem solver. Also note that special care must be taken with respect to the integration of the second equation in (19), in order to assure that the solution for R is a proper rotation matrix. Technically thus corresponds to a geometric inregration in SO(3).

2.1.4 Multiple shooting

Small changes in initial conditions and periodicity in the state might lead to convergence issues of single shooting. An alternative method, somehow in between single shooting and the method described in section 2.1.1 is to divide

the time interval in chunks, having separate simulations for each one of them. Let each one of these chunks, say M in number, be defined by an interval $[t_{c(i)}; t_{c(i+1)}]$ where $c(\cdot)$ defines the time indices deliming these chunks, such that

$$\begin{aligned} &c(0) = 0 \\ &c(M) = N \\ &c(0) < c(1) < \cdots < c(M) \end{aligned} \tag{22}$$

The resulting optimization problem is then

$$\begin{aligned} \text{minimize} \quad & \sum_{i=0}^{M-1} \sum_{k\,:\,c(i)\leq k<c(i+1)} \psi\left(R_k, \Phi(R_{c(i)}; t_{c(i)}, t_k)\right) \\ \text{w.r.t.} \quad & L, c, A_{1M} \\ \text{s.t.} \quad & \|A_{1M}\|^2 = 1 \end{aligned} \tag{23}$$

2.1.5 Variational integration approach

The usual approach for obtaining a discrete time formulation of a mechanical system is to discretize the continuous time dynamical system. That was the approach taken in previous sections, having Newton-Euler rigid body equations as a starting point. However, from a theoretical point of view, these equations are not fundamental and can be derived using a Lagrangian approach. That is, one writes the Lagrangian of a system, $L = T - U$, where T is the kinetic energy and U the potential energy. Then, one takes the action, which is the integral of L over a period of time, and the Newton-Euler equations are satisfied by the trajectory that results from minimizing the action over all possible trajectories (this is done classically using calculus of variations). This is all done in continuous time. The state variables are usually denoted as the pair $(q, \dot{q})$ where q is the generalized coordinates vector and $\dot{q}$ its time derivative.

In other words, the previous approach can be explained as one taking the continous time Lagrangian $L(q, \dot{q})$, minimizing the action to obtain the continuous time Newton-Euler equations, e.g. (1) in our case, and then discretize them to obtain a disrete time dynamical system.

However, there is an alternative approach, which essentially does the time discretization of the Lagrangian, and then follows in discrete time until obtaining a discrete time system. The discrete time Lagrangian is the integral of the continuous time Lagrangian over each discrete time step. Then the discrete time action is a summation, instead of an integral, and its minimization over all possible trajectories (now discrete time) results in a discrete time algebraic dynamical system.

While the continuous time Lagrangian is a function of the state, $L(q, \dot{q})$, the discrete time one is a function of two consecutive generalized coordinates, $L(q_k, q_{k+1})$. This non dependence on velocities also make this approach attractive, given that velocity estimation is often noisier. Informally, one can say that the pair (q_k, q_{k+1}) replaces the velocity.

A major advantage of this approach is that, while the discretization of the continuous time Newton-Euler equations result in a discrete time system that does not preserve neither energy nor linear/angular momentum, the *variational integration* approach[3] does (for conservative systems, that is, without dissipation). Thus, it is provides a more accurate discretization, for the same time steps. This has been explored by us recently[4], so the idea here is to apply the approach to our problem.

Following ([5]), which applies the variational integration approach to the attitude dynamics of a rigid body, the discrete time attitude system is expressed by:

$$\begin{aligned} \left\{F_k + F_{k+1}^T, J_d\right\} &= h^2 S(M_{k+1}) \\ R_{k+1} &= R_k F_k \end{aligned} \tag{24}$$

where the brackets is a binary matrix operator defined by $\{A, B\} = A^T B - B^T A$, R_k and R_{k+1} are the rotation matrices of two consecutive time steps, being F_k the incremental rotation between them (defined by the second equation), M_{k+1} is the torque at time step $k+1$, and h is the time step (as before). Matrix J_d is a non-standard representation of the moment of inertia, relating to J_B by the expression

$$J_B = \mathrm{tr}(J_d) I_3 - J_d \tag{25}$$

The torque M has two components, one resulting from the offset center of mass and another from the actuation. From (11) one can infer this torque to be expressed by

$$M = mS(c)R^T g + mA_{1M} u \tag{26}$$

recalling that A_{1M} is normalized to m. Define the operator $\bar{S}(\cdot)$ to be the inverse operation of $S(\cdot)$, recovering the vector v from a skew-symmetric matrix $S(v)$, that is, $\bar{S}(S(v)) = v$. Replacing this in (24), while noting that

[3]This is also known as discrete mechanics, and is an instance of a more general family of integrators called symplectic integrators or geometric integrators.

[4]Monica Ekal and Rodrigo Ventura. A dual quaternion based discrete variational approach for accurate and online inertial parameter estimation in free-flying robots. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA-20), pages 6021–6027, 2020.

[5]Lee et al...

both sides of the first equation of (24) are skew-symmetric matrices, we get

$$\begin{aligned} \bar{S}\left(\left\{F_k + F_{k+1}^T, J_{1d}\right\}\right) &= h^2 S(c) R_{k+1}^T g + h^2 A_{1M} u_{k+1} \\ R_{k+1} &= R_k F_k \end{aligned} \tag{27}$$

where $J_{1d} = J_d/m$.

Consider now that we have collected a sequence of samples $\{(R_k, u_k)\}$. Using the second equation of (29) we can obtain the values of $\{F_k\}$ using $F_k = R_k^T R_{k+1}$. The optimization problem to estimate the inertial parameters follows from the first equation of (29)

$$\begin{aligned} \text{minimize} \quad & \sum_k \left\| \bar{S}\left(\left\{F_k + F_{k+1}^T, J_{1d}\right\}\right) - h^2 S(c) R_{k+1}^T g - h^2 A_{1M} \right\|^2 \\ \text{w.r.t.} \quad & L, c, A_{1M} \\ \text{s.t.} \quad & \|A_{1M}\|^2 = 1 \end{aligned} \tag{28}$$

The standard moment of inertia J_B/m can be recovered from J_{1d} using (25).

2.2 Linear algebraic approach

Note that the cost function of (12), using $J_1 = LL^T$,

$$\sum_k \left\| J_1 \dot{\omega}_k + S(\omega_k) J_1 \omega_k - S(c) R_k^T g - A_{1M} u_k \right\|^2 \tag{29}$$

is a sum of squared norms, each one of them a linear function of the parameters $\{J_1, c, A_{1M}\}$. First, let us place all parameters into a parameter vector θ as

$$\theta = [j_{11}, j_{12}, j_{13}, j_{22}, j_{23}, j_{33}, c_1, c_2, c_3, a_{11}, \ldots, a_{36}]^T \tag{30}$$

where

$$J_1 = \begin{bmatrix} j_{11} & j_{12} & j_{13} \\ j_{12} & j_{22} & j_{23} \\ j_{13} & j_{23} & j_{33} \end{bmatrix} \qquad c = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \tag{31}$$

and $\{a_{ij}\}$ are the entries of matrix A_{1M} with the standard notation. Using this vector, the expression (29) can be written as

$$\sum_k \|\Phi_k \theta\|^2 \tag{32}$$

where each Φ_k has the form

$$\Phi_k = \left[\; \Phi_k^J \mid \Phi_k^c \mid \Phi_k^A \; \right] \tag{33}$$

defined by

$$\Phi_k^J = \begin{bmatrix} \dot{\omega}_x & \dot{\omega}_y - \omega_z\omega_x & \dot{\omega}_z + \omega_y\omega_x & -\omega_z\omega_y & \omega_y^2 - \omega_z^2 & \omega_y\omega_z \\ \omega_z\omega_x & \dot{\omega}_x + \omega_z\omega_y & \omega_z^2 - \omega_x^2 & \dot{\omega}_y & \dot{\omega}_z - \omega_x\omega_y & -\omega_x\omega_z \\ -\omega_y\omega_x & \omega_x^2 - \omega_y^2 & \dot{\omega}_x - \omega_y\omega_z & \omega_x\omega_y & \dot{\omega}_y + \omega_x\omega_z & \dot{\omega}_z \end{bmatrix} \tag{34}$$

for $\dot{\omega}_k = [\dot{\omega}_x\ \dot{\omega}_y\ \dot{\omega}_z]^T$ and $\omega_k = [\omega_x\ \omega_y\ \omega_z]^T$, by $\Phi_k^c = S(r)$ where $r = R_k^T g$ (since $-S(c)R_k^T g = -S(c)r = S(r)c$), and by

$$\Phi_k^A = -\left[\begin{array}{c|c|c} u_k^T & 0 & 0 \\ 0 & u_k^T & 0 \\ 0 & 0 & u_k^T \end{array}\right] \tag{35}$$

With this in mind, the cost function (32) can be developed in the following way

$$\begin{aligned} \sum_k \|\Phi_k\theta\|^2 &= \theta^T \left(\sum_k \Phi_k^T\Phi_k\right)\theta \\ &= \theta^T H\theta \end{aligned} \tag{36}$$

where $H = \sum_k \Phi_k^T\Phi_k$, thus a semi-definite positive (SDP) and symmetric matrix by construction.

The cost function of the optimization problem (12) is therefore equivalent to a quadratic form, where $\theta = 0$ is a trivial solution. Since the values to be estimated cannot be zero, we shall look for the direction of θ where the increase of this cost function is minimal. Ideally, in the absense of noise, the true parameters values would make the cost function zero.

Taking the spectral decomposition of H,

$$H = Q\Lambda Q^T \tag{37}$$

where Q is an orthonormal matrix containing the eigenvectors as columns, and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_{27})$ is a diagonal matrix of eigenvalues, all non-negative, since H is both SDP and symmetric. Then, the quadratic form (36) can be re-written as

$$\begin{aligned} \theta^T H\theta &= (\theta^Q)^T\Lambda\theta^Q \\ &= \sum_i \lambda_i(\theta_i^Q)^2 \end{aligned} \tag{38}$$

where $\theta^Q = [\theta_1^Q, \ldots, \theta_{27}^Q]^T = Q^T\theta$ is the projection of the parameters θ onto the orthonormal basis defined by the eigenvectors of H. From this, one can conclude that the direction of θ which yields the lowest increse in the cost function is the direction along the eigenvector corresponding to the lowest eigenvalue of H.

3 Gravity compensation

3.1 Attitude subsystem

The goal here is to produce an actuation input u that compensates for gravity effect on the center of mass c. In other words, we look for compensation signal s such that the dynamics of (11) is equivalent to a zero gravity dynamics, that is

$$J_1\dot{\omega} + S(\omega)J_1\omega = A_{1M}u_0 \tag{39}$$

where u_0 is the new input signal, and recalling that $J_1 = LL^T$. Let $u = u_0 + s$ and s a signal satisfying

$$A_{1M}s + S(c)R^T g = 0 \tag{40}$$

then (11) becomes equivalent to (39). Since $A_{1M} \in \mathbb{R}^{3\times 6}$, the linear system (40) is underspecified, but if we consider that this compensation should not produce any force, $F = 0$, we have $A_{1F}s = 0$, and thus

$$A_1 s = \begin{pmatrix} 0 \\ -S(c)R^T g \end{pmatrix} \tag{41}$$

Assuming A_1 full rank[6], this equation has a unique solution given by

$$s = A_1^{-1}\begin{pmatrix} 0 \\ -S(c)R^T g \end{pmatrix} \tag{42}$$

3.2 Full system

Let us extend the above results to the whole free-flyer dynamics. As before, we seek an actuation s such that the dynamics (7) with $u = s + u_0$ becomes equivalent to

$$\begin{pmatrix} R^T\dot{v} \\ J_1\dot{\omega} + S(\omega)J_1\omega \end{pmatrix} = A_1 u_0 \tag{43}$$

To do so, the signal s must satisfy this linear system of equations

$$\begin{pmatrix} R^T\left[S^T(c)\dot{\omega} + S^2(\omega)c\right] \\ S(c)v - S(c)R^T g \end{pmatrix} = A_1 s \tag{44}$$

from which this signal can be easly obtained from

$$s = A_1^{-1}\begin{pmatrix} R^T\left[S^T(c)\dot{\omega} + S^2(\omega)c\right] \\ S(c)v - S(c)R^T g \end{pmatrix} \tag{45}$$

[6] It is so for the Space Cobot, by design.

A Rough estimation of Space CoBot inertial parameterts

If the a pendulum effect is strongly manifested in the Space CoBot, one simple way to get a rough estimate of its inertial parameters is to model it as a classical pendulum.

Consider a pendulum with mass m and length r. Its moment of inertia is given by

$$J_p = mr^2 \tag{46}$$

and its natural frequency is given by

$$\omega_p^2 = \frac{mGr}{J_p} \tag{47}$$

where $\omega_p = 2\pi/T_p$ for T_p being the oscillation period.

Let m be the mass of Space CoBot, measured, e.g., with a scale, and consider two simple experiments, performed without any actuation:

1. the robot is allowed free for a sufficient amount of time so that it becomes static, and take R_0 as the rotation matrix provided by the IMU;

2. the robot is manually perturbed to oscillate passively as a pendulum, and take ω_p to be its oscillation frequency[7].

Then, a rough estimation of its inertial parameters c and J can be easily obtained from:

$$\begin{aligned} r &= \frac{G}{\omega_p^2} \\ c &= rR_0^T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ J_B &= R_0 \begin{bmatrix} J_p & 0 & 0 \\ 0 & J_p & 0 \\ 0 & 0 & 0 \end{bmatrix} R_0^T \end{aligned} \tag{48}$$

where the first equation follows from replacing (46) in (47) and solving for r, and J_p is computed using (46).

[7]For instance, use a chronometer to measure n oscilations and derive the frequency from there, or measure directly from the IMU measurements over a period of time.